

Smooth Collapse of Trapped Void Fragments in an Incompressible Volume-of-Fluid Method

A working research note motivated by Truchas

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Abstract

In a one-fluid incompressible volume-of-fluid (VOF) calculation, a resolved void region provides a prescribed void-pressure boundary to the liquid. When a small void fragment becomes enclosed within one or a few mixed cells, that internal boundary is no longer represented. The mixed cells are then treated as incompressible and a nonzero residual void fraction may persist indefinitely. Truchas contains an experimental collapse model whose rate is tied to the magnitude of the recent local change in void fraction. This adapts collapse to the local interface-evolution time scale, but it switches itself off when the VOF field becomes stationary. Pressure-driven filling against a closed ceiling is a particularly clear failure case.

This note organizes several possible remedies. They include fixed-time relaxation, flow-time scaling, pressure-driven relaxation, Rayleigh-collapse-inspired laws, a compressible trapped-gas model, and a void-fraction-dependent pressure compliance. The last of these gives a useful synthesis: an enclosed mixed cell is treated as a continuously disappearing internal pressure boundary. It adds a nonnegative reaction term to the pressure projection, anchors a nearly void cell near the void pressure, and recovers ordinary incompressibility as the void fraction vanishes. The note derives the common kinematic framework, gives both absolute-pressure and incremental pressure-correction forms, and develops a capillary-biased signed variant. That variant allows a large unresolved pocket to expand or contract like open void, but increasingly biases a small remnant toward collapse through a subgrid Young–Laplace pressure shift. The note identifies unresolved choices and proposes a staged implementation and verification program.

Status. Revision 3 adds capillary-biased signed compliance and its Truchas incremental pressure-correction form in Sec. 6. A first signed-compliance prototype has performed well in limited testing; the capillary bias remains a candidate extension. This is a research note, not a specification. Equations labeled as candidate models are hypotheses to be evaluated. The reconstruction of the existing Truchas algorithm should be checked against the precise source revision used for any implementation work.

Contents

1	Problem and modeling scope	4
2	Kinematic framework: collapse as permitted volumetric strain	4
3	The existing Truchas model	5
3.1	Source-level reconstruction	5
3.2	Plausible motivation	5
3.3	Failure when the VOF field becomes stationary	5
4	Requirements for a successor model	6
5	Candidate collapse-rate models	6
5.1	Fixed-time relaxation	6
5.2	Resolved-flow time scales	7
5.3	Pressure-driven relaxation	7
5.4	Rayleigh-collapse-inspired inertial scaling	8
5.5	Actual compressible trapped gas	8
6	Pressure compliance as a disappearing internal boundary	9
6.1	Connection to a void-fraction rate law	9
6.2	Signed, one-sided, and capillary-biased response	10
6.3	Absolute-pressure and incremental pressure-correction forms	11
6.3.1	Absolute-pressure form	11
6.3.2	Incremental pressure-correction form	11
6.4	Endpoint treatment and discrete regularization	12
7	Eligibility: which void is allowed to collapse?	13
8	Discrete algorithms worth testing	13
8.1	Algorithm A: lagged rate source	13
8.2	Algorithm B: implicit linear compliance	14
8.3	Algorithm C: one-sided or nonlinear implicit response	14
9	Conservation, stability, and interpretation	14
9.1	Liquid-volume conservation	14

9.2	What is and is not physical	14
9.3	Pressure sign, interface motion, and cavitation	15
10	Verification and discriminating tests	15
10.1	Local and one-dimensional tests	15
10.2	Two- and three-dimensional tests	15
10.3	Convergence matrix and diagnostics	16
11	Recommended research sequence	16
12	Open questions	17
13	Conclusion	17
A	Dimensions and notation	18
B	Minimal pseudocode for incremental implicit compliance	18

1 Problem and modeling scope

Let F denote the volume fraction of real incompressible fluid in a VOF description [3], and let α denote void fraction. For now assume that no solid fraction occupies the cell, so

$$F + \alpha = 1. \quad (1)$$

A completely void cell adjacent to liquid acts as part of the free surface and supplies a prescribed pressure, denoted p_v (often zero gauge pressure), to the liquid pressure solve. Once void is enclosed in mixed cells and no cell in the fragment is treated as pure void, the discrete internal $p = p_v$ boundary disappears. Ordinary projection then makes the mixed region incompressible. The remaining void behaves numerically like an incompressible constituent and can persist even though the application intends void to be an absence of material rather than a conserved gas phase.

The immediate objective is therefore not a complete physical model of gas bubbles. It is a subgrid model that permits an unresolved, enclosed void volume to be replaced by liquid through actual liquid flux. The model should not act on the exterior-connected free surface and should turn itself off continuously when the void has vanished.

Two distinct goals must remain separate:

1. A *numerical void-annihilation model* intentionally permits enclosed void volume to reach zero.
2. A *compressible trapped-gas model* conserves gas mass. Compression changes its volume but, without dissolution, condensation, or venting, does not make it disappear.

The first goal is the one addressed primarily here.

2 Kinematic framework: collapse as permitted volumetric strain

Conservation of incompressible real fluid is

$$\frac{\partial F}{\partial t} + \nabla \cdot (F \mathbf{u}) = 0. \quad (2)$$

Expanding the divergence and using $F = 1 - \alpha$ gives

$$\frac{D\alpha}{Dt} = F \nabla \cdot \mathbf{u} = (1 - \alpha) \nabla \cdot \mathbf{u}. \quad (3)$$

Suppose an eligible trapped fragment is assigned a nonnegative void-destruction rate R :

$$\frac{D\alpha}{Dt} = -R(\alpha, p, \mathbf{u}, \dots), \quad R \geq 0. \quad (4)$$

Consistency with Eq. (2) requires the modified divergence constraint

$$\boxed{\nabla \cdot \mathbf{u} = -\frac{R}{1 - \alpha}.} \quad (5)$$

Thus trapped-void collapse is naturally formulated as a localized negative volumetric strain. Liquid replacing the lost void must enter through the velocity fluxes; the volume fraction is not simply edited after the flow solve. Equation (5) is the common framework for all rate laws considered below.

The factor $(1 - \alpha)^{-1}$ is kinematically required but becomes singular in the pure-void limit. That endpoint is already governed by the pure-void pressure boundary, so Eq. (5) is intended for mixed cells. A discrete method must join these regimes carefully.

3 The existing Truchas model

The Truchas manual describes an experimental model intended to eliminate persistent void fragments that VOF otherwise treats as incompressible. It uses a dimensionless relaxation factor in $[0, 1]$, with default 0.1, and describes the factor as roughly the inverse of the number of time steps over which inertial forces are allowed to collapse a cell's void [1]. The model is implemented in the pressure projection, which is the appropriate algorithmic location for enforcing a nonzero divergence [4].

3.1 Source-level reconstruction

The source-level behavior discussed here can be summarized as follows. Define the ordinary pre-collapse VOF change

$$\Delta\alpha = \alpha^{n+1,*} - \alpha^n. \quad (6)$$

For an enclosed mixed cell, the collapse amount made available to the projection is, schematically,

$$\boxed{\delta\alpha_c = \min(|\Delta\alpha|, r\alpha)}, \quad (7)$$

where r is `void_collapse_relaxation`. Both signs of $\Delta\alpha$ contribute the same collapse sign; consequently the method does not merely continue an already shrinking fragment. It uses $|\Delta\alpha|$ as a sensor for the rate at which the local VOF configuration is evolving. The exact algebra and candidate-cell test should be cited to a recorded Truchas commit before this note is circulated as a code description [2].

3.2 Plausible motivation

When the cap in Eq. (7) is inactive,

$$\frac{\delta\alpha_c}{\Delta t} \simeq \left| \frac{D\alpha}{Dt} \right|. \quad (8)$$

The implied local VOF-evolution time is

$$\tau_\alpha = \frac{\alpha}{|D\alpha/Dt|}. \quad (9)$$

The collapse therefore proceeds on approximately the same intrinsic time scale as the resolved interface evolution. For an interface sweeping across cells, $|D\alpha/Dt| \sim U/h$, so the scale is related to a cell-crossing time. This is a clever heuristic: it avoids prescribing an independent dimensional collapse time, responds rapidly in violent filling and splashing, and is approximately invariant to time-step refinement while the uncapped estimate is valid. The parameter r then acts mainly as a per-step robustness limiter.

3.3 Failure when the VOF field becomes stationary

The same sensor creates the central defect:

$$|\Delta\alpha| \longrightarrow 0 \implies \delta\alpha_c \longrightarrow 0, \quad (10)$$

even when $\alpha > 0$ and the fragment remains enclosed. The model is triggered by recent kinematic evolution, not by the present existence of trapped void or by the force driving its collapse.

A pressure-fed mold filling against a closed horizontal ceiling is an almost diagnostic counterexample. Once the advancing liquid reaches the ceiling, the interface stops moving through the grid and the local VOF field approaches a stationary state. The remaining top-layer void can still be substantial, and the liquid pressure may remain positive, but Eq. (7) switches off. Ordinary incompressibility then prevents further net inflow and freezes the residual void.

There is a second conceptual weakness. A translating or deforming fragment can produce positive $\Delta\alpha$ in some cells and negative $\Delta\alpha$ in others without changing its total void volume. Local $|\Delta\alpha|$ is therefore not, in general, a measure of component collapse.

4 Requirements for a successor model

A useful replacement should satisfy the following requirements.

1. **Persistent activation:** an eligible cell with $\alpha > 0$ must not become inactive merely because $\Delta\alpha = 0$.
2. **Flux consistency:** lost void is replaced by liquid through the velocity field, using Eq. (5).
3. **Free-surface protection:** exterior-connected void must remain an ordinary free surface rather than being annihilated.
4. **Correct endpoints:** the pure-void limit behaves like $p = p_v$, while the zero-void limit recovers $\nabla \cdot \mathbf{u} = 0$.
5. **Pressure response:** positive pressure loading should continue to drive collapse in a stagnating pocket.
6. **Time-step convergence:** parameters should define a continuous-time law; per-step bounds may remain as numerical limiters.
7. **Controlled grid dependence:** any dependence on cell size or inferred fragment size should be explicit and testable.
8. **Solver compatibility:** the pressure operator should preferably remain coercive, or any nonlinearity should admit a robust active-set or fixed-point treatment.

5 Candidate collapse-rate models

Table 1 previews the alternatives. None of the constitutive laws by itself settles the separate question of which void is eligible for collapse.

5.1 Fixed-time relaxation

The simplest persistent law is

$$R = \frac{\alpha}{\tau_c}, \quad \nabla \cdot \mathbf{u} = -\frac{\alpha}{(1 - \alpha)\tau_c}. \quad (11)$$

It gives exponential decay along a trajectory and remains active until the void becomes negligible. A per-step bound such as $\delta\alpha_c \leq r_{\max}\alpha$ can be retained for robustness, but should not define the constitutive law. This model is an excellent baseline because its time-step convergence is easy to verify. Its weakness is the new dimensional parameter and its indifference to pressure: it collapses eligible void at the same rate whether weakly or strongly loaded.

Table 1: Candidate laws for $D\alpha/Dt = -R$.

Model	Rate R	Principal consequence
Existing Truchas	$\min(\Delta\alpha , r\alpha)/\Delta t$	Flow-responsive and simple, but becomes identically inactive for stationary trapped void.
Fixed time	α/τ_c	Persistent exponential decay with a clear continuous-time parameter; no pressure or local-dynamics response.
Local flow time	$C_u\alpha \mathbf{u} /h$ or $C_D\alpha\ \mathbf{D}\ $	Uses resolved dynamics without a dimensional time, but either reacts to translation or fails in pressure-loaded stagnation.
Pressure relaxation	$\alpha g((p - p_v)/p_0)/\tau_0$	Continues under static pressure loading; linear or rate-saturated variants are easy to construct.
Rayleigh-inspired	$K\alpha^{2/3}\sqrt{[p - p_v]_+}/\rho/h$ in 3-D	Inertial, pressure- and size-responsive, and reaches zero in finite time; introduces geometric and grid/component-size assumptions.
Compressible gas	EOS plus gas-mass conservation	Physically interpretable pressure-volume response, but gas does not disappear without additional mass-transfer physics.
Capillary-biased compliance	$\nabla \cdot \mathbf{u} = -C(\alpha)[p - p_v + \Pi_\sigma(\alpha, h)]$	Retains reversible open-void behavior for large fragments while shifting small remnants toward collapse; remains linear when the capillary shift is lagged.

Choosing $\tau_c = N_c\Delta t$ reproduces a fraction-per-step behavior but forfeits a continuous-time model. It may be useful only as a temporary compatibility mode.

5.2 Resolved-flow time scales

The apparent intent of the existing method can be preserved by replacing the vanishing VOF-change sensor with a nonzero local flow scale, for example

$$R_u = C_u\alpha\frac{|\mathbf{u}|}{h}, \quad (12)$$

$$R_D = C_D\alpha\|\mathbf{D}(\mathbf{u})\|, \quad \mathbf{D}(\mathbf{u}) = \frac{1}{2}(\nabla\mathbf{u} + \nabla\mathbf{u}^T). \quad (13)$$

The velocity scale collapses a fragment that is merely translating, even if it is not being compressed, and it vanishes at the stagnating ceiling. The strain-rate scale avoids sensitivity to rigid translation, but it too can vanish in a pressure-loaded stagnant pocket. These scales may be useful modifiers or fallback estimates, but neither addresses the motivating failure by itself.

5.3 Pressure-driven relaxation

A simple pressure-responsive family is

$$R = \frac{\alpha}{\tau_0}g\left(\frac{p - p_v}{p_0}\right), \quad (14)$$

where $g(x) \geq 0$ for $x > 0$. Two useful choices are

$$g(x) = [x]_+, \quad \text{linear pressure response,} \quad (15)$$

$$g(x) = \min([x]_+, 1), \quad \text{saturated response.} \quad (16)$$

The linear choice is equivalently $R = M\alpha[p - p_v]_+$, with $M = (\tau_0 p_0)^{-1}$. The saturated form defines τ_0 as the fastest relaxation time and prevents the rate from becoming arbitrarily large at high pressure. Both remain active in a stationary, pressure-loaded ceiling pocket. They introduce a pressure scale and a collapse time, but those parameters have clear units and meanings.

The one-sided positive part suppresses void growth when $p < p_v$. Omitting it produces a reversible compliance model that can grow void under tension, which would amount to a rudimentary cavitation model and should be a separate decision.

5.4 Rayleigh-collapse-inspired inertial scaling

For an empty spherical cavity of initial radius R_0 in an infinite inviscid liquid under constant pressure difference $\Delta p > 0$, Rayleigh's collapse time is

$$t_R \simeq 0.915 R_0 \sqrt{\frac{\rho}{\Delta p}} \quad (17)$$

[5, 6]. This suggests the intrinsic scale

$$\tau_p = C_R L \sqrt{\frac{\rho}{[p - p_v]_+}}. \quad (18)$$

Taking $L = h$ yields a deliberately cell-scale model. Taking L as the radius of an entire connected component is more physically meaningful but requires component identification and distributed reductions.

At the cell level in three dimensions, an effective spherical radius is

$$R_{\text{eff}} = \left(\frac{3\alpha V_c}{4\pi} \right)^{1/3}. \quad (19)$$

Combining $R \sim \alpha/\tau_p$ with this radius gives the scaling

$$R_{\text{inertial}} = K_3 \frac{\sqrt{[p - p_v]_+ / \rho}}{V_c^{1/3}} \alpha^{2/3}. \quad (20)$$

In two dimensions the analogous exponent is $1/2$. Because $\dot{\alpha} \propto -\alpha^{2/3}$, the three-dimensional model reaches zero in finite time. The coefficient must be calibrated carefully: using the instantaneous Rayleigh time as α/R makes the total extinction time three times the initial local time, so matching Eq. (17) requires the corresponding factor of three.

Equation (20) is not the Rayleigh–Plesset equation. It borrows its pressure, density, length, and finite-extinction scaling for a subgrid closure. Cell geometry, walls, viscosity, surface tension, nonsphericity, and pressure variation are all omitted. It is therefore a second-generation candidate, not the first model to implement.

5.5 Actual compressible trapped gas

If the void is physically entrapped air, a more faithful model conserves gas mass and uses an equation of state, for example

$$p_g V_g^\gamma = \text{constant}. \quad (21)$$

Li and Carrica tightly couple bubble density, void fraction, pressure, and velocity in a two-phase solver with incompressible liquid and compressible bubbles [7]. Shin uses a polytropic gas law and represents gas compressibility in the source term of a unified pressure Poisson equation [8]. These works support the general numerical idea of coupling pressure and void-volume response implicitly.

They do not, however, solve the present annihilation problem: conserved gas can become highly compressed but cannot vanish. Dissolution, condensation, leakage, or venting would be required. A compressible-gas model should therefore be treated as a distinct future physics capability, not as a more elaborate implementation of numerical cleanup.

6 Pressure compliance as a disappearing internal boundary

The most useful synthesis is to interpret an unresolved trapped fragment as an internal $p = p_v$ boundary that should disappear continuously as $\alpha \rightarrow 0$. Introduce a nonnegative pressure compliance $C(\alpha, p)$ through

$$\boxed{\nabla \cdot \mathbf{u} = -C(\alpha, p) (p - p_v).} \quad (22)$$

For a one-sided annihilation law, replace $p - p_v$ by $[p - p_v]_+$.

The desired limits are

$$C \longrightarrow \infty \quad \text{as} \quad \alpha \rightarrow 1^-, \quad C \longrightarrow 0 \quad \text{as} \quad \alpha \rightarrow 0^+. \quad (23)$$

A large compliance strongly anchors the pressure near p_v while permitting rapid net inflow. A vanishing compliance recovers ordinary incompressibility and lets the pressure approach that of the surrounding liquid. More precisely, the model provides a large *capacity* for negative divergence near the void limit; the actual divergence is determined by the coupled global pressure solve.

The qualitative feedback loop is

$$\begin{aligned} \text{enclosed void} &\Rightarrow \text{pressure compliance} \Rightarrow \text{negative divergence} \\ &\Rightarrow \text{liquid inflow} \Rightarrow \text{less void and less compliance.} \end{aligned}$$

This loop does not require an antecedent $\Delta\alpha$.

6.1 Connection to a void-fraction rate law

If

$$\frac{D\alpha}{Dt} = -\lambda(\alpha) [p - p_v]_+, \quad (24)$$

then Eq. (3) gives

$$C(\alpha) = \frac{\lambda(\alpha)}{1 - \alpha}. \quad (25)$$

The linear pressure-relaxation model is obtained with

$$\lambda(\alpha) = \frac{\alpha}{\tau_0 p_0}, \quad \boxed{C(\alpha) = \frac{\alpha}{(1 - \alpha)\tau_0 p_0}.} \quad (26)$$

This choice simultaneously gives exponential void decay at fixed pressure, strong anchoring near the void endpoint, and exact recovery of incompressibility as $\alpha \rightarrow 0$.

More generally,

$$C(\alpha) = C_0 \frac{\alpha^m}{(1-\alpha)^n}, \quad m > 0, n > 0, \quad (27)$$

allows the two endpoint transitions to be shaped independently. There is little reason to introduce both exponents before the simplest choice $m = n = 1$ has been tested.

6.2 Signed, one-sided, and capillary-biased response

The symmetric form of Eq. (22),

$$\nabla \cdot \mathbf{u} = -C(\alpha)(p - p_v), \quad (28)$$

allows both signs of divergence. It collapses an existing fragment when $p > p_v$ and expands it when $p < p_v$. By contrast,

$$\nabla \cdot \mathbf{u} = -C(\alpha) [p - p_v]_+ \quad (29)$$

is an irreversible annihilation law. The signed law is linear in pressure and a first prototype has worked well in limited tests. Moreover, positive divergence in a cell that is mostly void need not be interpreted as nucleating a new cavity: it can represent motion of an already existing unresolved liquid–void interface. If $C(0) = 0$, a completely liquid cell cannot nucleate void through this closure alone.

A compromise is to retain signed compliance but shift its equilibrium pressure by a capillary term:

$$\boxed{\nabla \cdot \mathbf{u} = -C(\alpha) [p - p_v + \Pi_\sigma(\alpha, h)] = -C(\alpha) [p - p_{\text{eq}}]}, \quad p_{\text{eq}} = p_v - \Pi_\sigma. \quad (30)$$

Thus the fragment collapses for $p > p_v - \Pi_\sigma$, is in equilibrium at $p = p_v - \Pi_\sigma$, and expands only when the liquid pressure falls below that threshold. For a large, nearly open void, Π_σ should tend to zero so that both signs remain available around p_v . As the unresolved fragment becomes small, Π_σ increases and progressively suppresses expansion.

The sign follows the Young–Laplace jump. For a spherical void in three dimensions,

$$p_v - p_\ell = \frac{2\sigma}{R_{\text{eff}}}, \quad (31)$$

so the liquid-side equilibrium pressure is below the void pressure. More generally the curvature scale is $(d-1)/R_{\text{eff}}$ in d spatial dimensions [9]. A deliberately cell-scale closure is

$$\boxed{\Pi_\sigma(\alpha, h) = c_\sigma \frac{\sigma}{h} B(\alpha)}, \quad h = \begin{cases} V_c^{1/3}, & \text{three dimensions,} \\ A_c^{1/2}, & \text{two dimensions,} \end{cases} \quad (32)$$

where σ is the physical liquid–void surface tension, c_σ is a dimensionless subgrid curvature/geometry coefficient, and $B(\alpha)$ controls when the bias turns on. A fixed, documented B gives this part of the model two inputs, σ and c_σ . If σ is already a material property, only c_σ is a new calibration parameter. If it is not independently known, the projection identifies only their product $\Sigma_{\text{eff}} = c_\sigma \sigma$. Any adjustable transition point or exponent inside B would add parameters and should be avoided initially.

A parameter-free baseline shape that satisfies the required open-void limit is

$$\boxed{B(\alpha) = (1 - \alpha)^2}. \quad (33)$$

It gives $B(0) = 1$, $B(1) = 0$, and, together with Eq. (26), $CB \sim \alpha(1 - \alpha)$. A more literal effective-radius model could multiply this fade by $\alpha^{-1/d}$, but that introduces a singular endpoint and a regularization choice. The bounded quadratic shape is the cleaner first test.

The limiting behavior of the *product* $C\Pi_\sigma$, not just Π_σ , must be checked. Since Eq. (26) has $C \sim (1 - \alpha)^{-1}$ as $\alpha \rightarrow 1$, recovering an unbiased open-void limit requires $B(\alpha) = o(1 - \alpha)$ there, or an explicit cap/handoff must keep $C\Pi_\sigma$ bounded and vanishing. The closure should be applied only to isolated, underresolved fragments. If resolved surface tension already supplies the relevant pressure jump, the subgrid term must be blended or disabled to avoid double counting.

6.3 Absolute-pressure and incremental pressure-correction forms

The compliance model is compatible with either a projection that solves for the new pressure or an incremental method that uses an existing pressure in the predictor and solves for a correction. The distinction changes the algebraic right-hand side, not the underlying model.

6.3.1 Absolute-pressure form

Let $\tilde{\mathbf{u}}$ be a provisional velocity that has not yet received the new pressure force, and write

$$\mathbf{u}^{n+1} = \tilde{\mathbf{u}} - \mathbf{K}\nabla p^{n+1}, \quad \mathbf{K} = \Delta t \rho^{-1} \mathbf{I} \quad (34)$$

for the simplest scalar-density projection. Here $\mathbf{K}$ may instead denote the actual discrete face mobility used by the code. With compliance lagged in α , substitution into Eq. (22) gives

$$\boxed{-\nabla \cdot (\mathbf{K}\nabla p^{n+1}) + C(\alpha^*)p^{n+1} = -\nabla \cdot \tilde{\mathbf{u}} + C(\alpha^*)p_v.} \quad (35)$$

Thus an eligible mixed cell receives a nonnegative diagonal reaction term. Equation (35) is a smooth penalty or Robin-like transition: large C approximates a void-pressure condition, intermediate C partially anchors pressure, and $C = 0$ gives the ordinary interior projection. The added reaction term is coercive and should improve rather than damage the definiteness of the linear operator, although strong coefficient variation can still affect conditioning.

For capillary-biased signed compliance, replace p_v by $p_{\text{eq}} = p_v - \Pi_\sigma$. With α , h , and Π_σ lagged, the absolute-pressure equation remains linear:

$$-\nabla \cdot (\mathbf{K}\nabla p^{n+1}) + Cp^{n+1} = -\nabla \cdot \tilde{\mathbf{u}} + C(p_v - \Pi_\sigma). \quad (36)$$

6.3.2 Incremental pressure-correction form

For Truchas, let p^n be the existing pressure used in the predictor. The incremental method can be written

$$\mathbf{u}^* = \tilde{\mathbf{u}} - \mathbf{K}\nabla p^n, \quad (37)$$

$$\mathbf{u}^{n+1} = \mathbf{u}^* - \mathbf{K}\nabla \phi, \quad (38)$$

$$p^{n+1} = p^n + \phi, \quad (39)$$

where ϕ is the pressure correction. Requiring the corrected velocity to satisfy

$$\nabla \cdot \mathbf{u}^{n+1} = -C(\alpha^*)(p^{n+1} - p_v) \quad (40)$$

gives the incremental analog of Eq. (35):

$$\boxed{-\nabla \cdot (\mathbf{K} \nabla \phi) + C(\alpha^*)\phi = -\nabla \cdot \mathbf{u}^* - C(\alpha^*)(p^n - p_v)}. \quad (41)$$

This is the form directly applicable to Truchas. The reaction coefficient multiplies the pressure *correction*, but the right-hand side must also contain the offset of the predictor pressure from the void pressure. Omitting that offset would impose $\nabla \cdot \mathbf{u}^{n+1} = -C\phi$, anchoring the correction to zero rather than anchoring the updated physical pressure to p_v .

Equations (35) and (41) are algebraically equivalent when the predictor and corrector use the same discrete gradient, divergence, mobility, and pressure boundary treatment. Substituting $\phi = p^{n+1} - p^n$ and Eq. (37) into Eq. (41) recovers Eq. (35). Incremental correction therefore does not obstruct the proposed compliance model; it simply splits the absolute-pressure equation around the pressure already used in the predictor.

The corresponding capillary-biased incremental equation is

$$\boxed{-\nabla \cdot (\mathbf{K} \nabla \phi) + C\phi = -\nabla \cdot \mathbf{u}^* - C[p^n - p_v + \Pi_\sigma(\alpha^*, h)]}. \quad (42)$$

Only the right-hand-side offset differs from Eq. (41); the matrix and its nonnegative reaction term are unchanged. Consequently the capillary-biased signed model preserves the linear pressure-correction solve when Π_σ is evaluated from lagged geometry.

If the code uses a scaled pressure update, for example $p^{n+1} = p^n + \omega_p \phi$, the reaction contribution multiplying ϕ becomes $\omega_p C\phi$. Likewise, $\mathbf{K}$ must be the actual coefficient relating the velocity correction to $\nabla \phi$, including any BDF or variable-density factors. These details should be taken from the Truchas discrete update rather than inferred from the schematic equations above.

Using $[p - p_v]_+$ makes the strict one-sided equation semilinear. A simple active-set method can classify cells from the trial updated pressure $p^n + \phi$, solve the resulting linear system, and repeat until the active set stops changing. In incremental form, with $A_i = 1$ on the active set and zero otherwise, Eq. (41) becomes

$$-\nabla \cdot (\mathbf{K} \nabla \phi) + AC\phi = -\nabla \cdot \mathbf{u}^* - AC(p^n - p_v). \quad (43)$$

A smooth approximation to the positive part is another option, but it introduces a regularization scale and permits some response for $p < p_v$.

6.4 Endpoint treatment and discrete regularization

The singular compliance in Eq. (26) is a meaningful limiting statement, not a number to evaluate at $F = 0$. Practical options include:

1. retain the existing pure-void Dirichlet treatment for $F \leq F_D$, and use compliance only for $F > F_D$;
2. cap C at a value that makes the reaction term dominate the local pressure stencil by a prescribed factor;

3. replace $1 - \alpha$ by $\max(1 - \alpha, F_\epsilon)$, while verifying that results are insensitive to F_ϵ ;
4. integrate the local linear pressure-relaxation law exactly over a step, reducing sensitivity to a large explicit collapse rate.

The first two preserve the interpretation most transparently. A smooth handoff can be constructed by choosing the cap to match the discrete Dirichlet penalty at F_D .

7 Eligibility: which void is allowed to collapse?

The rate law and the eligibility test are logically independent. The ideal criterion is topological:

$$\mathcal{T}_i = \begin{cases} 1, & \text{void in cell } i \text{ is disconnected from exterior void,} \\ 0, & \text{otherwise.} \end{cases} \quad (44)$$

Only cells with $\mathcal{T}_i = 1$ receive the modified divergence constraint.

The existing inexpensive criterion—a mixed cell with no neighboring completely void cell—captures many one-cell fragments but can misclassify a thin or underresolved free-surface filament. It also cannot identify a trapped component spanning several mixed cells when no member crosses the pure-void threshold.

A graph-based method would mark cells carrying appreciable void, connect face neighbors, seed components touching exterior void or a pressure outlet, and classify all unseeded components as trapped. In parallel this becomes a distributed flood fill or connected-component problem. It need not necessarily be run every step: connectivity could be updated only after material topology changes beyond a threshold. Hysteresis in the graph thresholds is desirable to prevent classification chatter.

Component identification also enables physically better length scales and diagnostics:

$$V_{v,k} = \sum_{i \in k} \alpha_i V_i, \quad R_k = \left(\frac{3V_{v,k}}{4\pi} \right)^{1/3}. \quad (45)$$

However, global connectivity should not be a prerequisite for testing a new constitutive law. A staged study can begin with the present local eligibility test and isolate the rate-law effect.

8 Discrete algorithms worth testing

8.1 Algorithm A: lagged rate source

This is the lowest-cost experiment.

1. Advect VOF to obtain α^* and compute the current eligibility mask.
2. Evaluate a rate $R(\alpha^*, p^n, \mathbf{u}^*)$.
3. Impose $\nabla \cdot \mathbf{u}^{n+1} = -R/(1 - \alpha^*)$ as a known source in the existing projection.
4. Limit the predicted removal to available void and, if needed, $\delta\alpha_c \leq r_{\max}\alpha^*$.
5. Update VOF using the same corrected fluxes used by continuity.

Algorithm A can compare fixed-time, saturated-pressure, and Rayleigh-inspired rates with minimal pressure-solver changes. Lagged pressure may delay the onset by one step and can be unstable for a strong pressure response.

8.2 Algorithm B: implicit linear compliance

Insert Eq. (26) directly into the projection to obtain Eq. (35), or its incremental Truchas form Eq. (41). Lag α , but solve updated pressure and collapse simultaneously. This is the cleanest first implementation of the disappearing-boundary idea. It should be compared against Algorithm A at weak coupling and tested for conditioning as $\alpha \rightarrow 1$.

The capillary-biased signed variant uses Eq. (42). It has the same matrix as the unbiased signed scheme and changes only the known pressure offset on the right-hand side. This makes it the most direct extension of the tested linear prototype.

After the pressure solve, compute the collapse volume from either the integrated corrected face fluxes or the cell divergence. These two evaluations should agree to the pressure solver tolerance. That comparison is an important implementation diagnostic.

8.3 Algorithm C: one-sided or nonlinear implicit response

Use either the active-set form

$$\nabla \cdot \mathbf{u} = -C(\alpha) [p - p_v]_+ \quad (46)$$

or an inertial response derived from Eq. (20). The pressure operator is then nonlinear. This algorithm is justified only after the linear-compliance version has demonstrated that pressure coupling cures the stationary-pocket failure.

9 Conservation, stability, and interpretation

9.1 Liquid-volume conservation

The purpose of Eq. (5) is to make the increase in liquid volume equal the net liquid inflow. In a finite-volume implementation, the same face fluxes must be used in the pressure correction and the VOF update. A post hoc decrement of α would manufacture liquid and should be avoided.

Truchas documentation reports that larger relaxation values in the experimental model can increase mass loss and that additional VOF subcycling improves the result [1]. This makes discrete flux consistency and subcycle coupling first-class acceptance criteria for any successor.

9.2 What is and is not physical

The pressure-compliance model is best described as a numerical closure for a missing subgrid pressure boundary. Unless its parameters are derived and validated for a specific gas-removal mechanism, it should not be presented as a physical bubble model. Rayleigh scaling provides a defensible inertial time scale, but a cell-scale nonspherical fragment adjacent to walls is far outside Rayleigh's assumptions.

The model is intentionally dissipative in the sense that it removes an unresolved degree of freedom. A useful study should monitor pressure work associated with $p \nabla \cdot \mathbf{u}$, because overly rapid annihilation may create spurious pressure transients or remove kinetic energy at a grid-dependent rate.

9.3 Pressure sign, interface motion, and cavitation

A symmetric linear compliance permits positive divergence when $p < p_v$. In an eligible mixed cell this can represent expansion of an existing unresolved interface, especially when the cell is mostly void; it is not automatically nucleation. Because $C(0) = 0$, the closure cannot create void from an exactly full liquid cell. It can, however, amplify a tiny numerical remnant under tension, so thresholds, topology, and pressure histories still require scrutiny.

The strict positive-part law forbids expansion but may make a mostly void cell behave less like the open void that it is replacing. Equation (30) offers an intermediate interpretation: reversible response near the large-void limit, with an increasingly negative expansion threshold as the remnant shrinks. It is not a complete cavitation model because it contains neither a nucleation criterion nor gas mass, thermodynamics, or bubble dynamics.

10 Verification and discriminating tests

The following hierarchy is designed to distinguish constitutive errors from topology and VOF-transport errors.

10.1 Local and one-dimensional tests

1. **Prescribed-pressure decay.** Hold $p - p_v$ fixed and verify the analytic $\alpha(t)$ for fixed-time and linear-pressure relaxation over a sequence of time steps.
2. **Finite-extinction law.** Verify that an $\alpha^{2/3}$ rate gives a linear decrease of $\alpha^{1/3}$ in time and the predicted extinction time.
3. **Reaction-Poisson cell.** In a one-dimensional column, verify the smooth transition from a strong p_v anchor to an ordinary interior pressure as α decreases.
4. **Flux identity.** Check that integrated net inflow equals the computed loss of void to solver tolerance.
5. **Signed capillary threshold.** Prescribe pressure on both sides of $p_v - \Pi_\sigma$ and verify the predicted sign of divergence, including zero response at the equilibrium threshold.

10.2 Two- and three-dimensional tests

1. **Pressure-fed ceiling fill.** This is the primary regression. Record the top-layer void volume, pressure, divergence, and inflow after macroscopic filling stops. The current model is expected to plateau; a successful pressure model should continue toward zero with pressure rising smoothly toward neighboring liquid pressure and divergence decreasing toward zero.
2. **Stationary enclosed fragment.** Initialize a trapped fragment with $\Delta\alpha = 0$ and positive surrounding pressure. This directly tests persistent activation.
3. **Translating fragment.** Advect an enclosed fragment at nearly uniform velocity with weak pressure loading. This exposes the spurious translation sensitivity of $h/|\mathbf{u}|$ scaling.
4. **Exterior free surface and thin filament.** Confirm that eligibility logic never collapses exterior-connected void, including underresolved connections near the classification threshold.
5. **Multi-cell component.** Compare local and graph-based eligibility and, for inertial laws, cell-size versus component-size length scales.

6. **Capillary-bias sweep.** Vary α , h , σ , and c_σ . Verify that large-void behavior approaches unbiased signed compliance, small remnants resist expansion, and the measured threshold scales as $c_\sigma\sigma/h$.

10.3 Convergence matrix and diagnostics

Each discriminating case should vary Δt , VOF subcycling, mesh spacing, collapse parameters, and pressure-solver tolerance. At minimum record

$$V_v(t) = \sum_i \alpha_i V_i, \quad \sum_i V_i \nabla \cdot \mathbf{u}_i, \quad p_{\min}, p_{\max}, \quad \text{liquid-volume error,} \quad \text{pressure iterations.}$$

For the ceiling-fill problem, cellwise histories of $(\alpha, p, \nabla \cdot \mathbf{u}, C)$ in the top layer are more informative than images alone.

11 Recommended research sequence

The models should be introduced in an order that makes failures interpretable.

1. **Document the baseline.** Record the exact Truchas commit, candidate-cell test, source formula, sign convention, and mapping between projection divergence and VOF correction. Reproduce the ceiling-fill plateau quantitatively.
2. **Implement a fixed-time source.** Use Eq. (11) with the current eligibility test. This establishes that persistent activation alone can eliminate the plateau and provides a time-step-convergence baseline.
3. **Implement lagged saturated pressure relaxation.** Use Eq. (14). Determine whether pressure response improves the temporal and spatial pattern of collapse.
4. **Characterize the signed-compliance prototype.** The reaction term in Eq. (41) has been implemented and has performed well in limited testing. Quantify pressure smoothness, conservation, conditioning, time-step dependence, and sensitivity to tiny residual voids.
5. **Add the capillary bias.** Implement Eq. (42) first with $B(\alpha) = (1 - \alpha)^2$. Compare it directly with unbiased signed compliance and the strict one-sided active-set form.
6. **Improve eligibility only after the rate law is understood.** Add graph connectivity and hysteresis, using cases that distinguish multi-cell trapped void from exterior-connected filaments.
7. **Evaluate inertial scaling.** Test Eq. (20) with both cell and component length scales. Retain it only if it reduces parameter and mesh sensitivity relative to the simpler pressure law.

The most promising next model is therefore

$$\boxed{\nabla \cdot \mathbf{u} = -\mathcal{T} \frac{\alpha}{(1 - \alpha)\tau_0 p_0} \left[p - p_v + c_\sigma \frac{\sigma}{h} B(\alpha) \right]}, \quad (47)$$

treated implicitly in pressure with α lagged and with a controlled pure-void handoff. The strict positive-part and unbiased signed laws are the essential comparisons. Equation (47) preserves the successful linear structure of the prototype, directly fixes the stationary-pocket failure, and embodies the desired transition from reversible open-void response to capillary-biased disappearance of a small remnant.

12 Open questions

The study should not hide the following unresolved choices.

1. What fixed form of $B(\alpha)$ provides the desired transition without adding poorly identifiable shape parameters?
2. Should σ be the physical material property with only c_σ calibrated, or should the model expose a single effective parameter Σ_{eff} ?
3. At what void fraction and pressure should positive divergence be permitted, and what safeguards prevent amplification of numerical remnants?
4. What pressure should be used for p_v : zero gauge or an imposed exterior pressure?
5. Can $\tau_0 p_0$ be related to a discrete inertial impedance, reducing the number of user parameters?
6. Should the compliance be singular as $F \rightarrow 0$, or merely large relative to the local pressure stencil?
7. Is a component-aware length scale necessary for mesh convergence, or is deliberate cell-scale annihilation the more honest numerical model?
8. How often must connectivity be recomputed in a distributed calculation, and what hysteresis thresholds prevent chatter without retaining false connections?
9. Does pressure-compliance work create unacceptable transients when a cell changes from exterior-connected to trapped?
10. How should collapse couple to VOF subcycling so that liquid volume remains conservative at both the flow-step and subcycle levels?

13 Conclusion

The trapped-void pathology is best understood as the loss of an internal void-pressure boundary when a fragment becomes underresolved. The existing Truchas model addresses the problem in the right place—the pressure projection—and plausibly tries to match the intrinsic local interface-evolution time. Its $|\Delta\alpha|$ sensor, however, vanishes precisely in the important case of a stationary pressure-loaded pocket.

All candidate models can be expressed as localized negative divergence consistent with real-fluid conservation. Fixed-time relaxation is the essential baseline; flow-time laws preserve part of the old motivation but fail in stagnation; pressure and Rayleigh laws retain a nonzero physical driver; and a true gas model answers a different question. The pressure-compliance formulation unifies the most useful features. It replaces an abrupt switch between $p = p_v$ and ordinary interior pressure with a reaction term whose strength decreases with void fraction. This is both a concrete algorithm and a clear interpretation: a continuously disappearing subgrid pressure boundary.

Limited tests of the linear signed formulation support that interpretation. A capillary-biased signed law is a natural refinement: it preserves reversible response for a mostly void cell, retains the same linear incremental pressure matrix, and shifts the expansion threshold downward as the fragment becomes small. Its value will depend on choosing and validating the subgrid curvature function without duplicating resolved surface-tension physics.

Symbol	Dimensions	Meaning
F	1	real-fluid volume fraction
α	1	void volume fraction
R	T^{-1}	material void-destruction rate
C	$P^{-1}T^{-1}$	pressure compliance in Eq. (22)
λ	$P^{-1}T^{-1}$	pressure-to-void-rate coefficient
p_v	P	nominal void pressure
p_{eq}	P	capillary-shifted liquid-side equilibrium pressure
Π_σ	P	subgrid capillary pressure shift
σ	PL	physical liquid-void surface tension
c_σ	1	subgrid curvature/geometry coefficient
$B(\alpha)$	1	prescribed void-fraction activation function for capillary bias
p_0	P	pressure scale for relaxation
τ_0	T	collapse relaxation time
h	L	representative cell length
V_c	L^3	cell volume in three dimensions

A Dimensions and notation

B Minimal pseudocode for incremental implicit compliance

```

alpha_star = advect_vof(...)
trapped    = classify_trapped_void(alpha_star, ...)

for each flow cell i:
    if trapped(i) and F(i) > F_D:
        C(i) = alpha_star(i) / (F(i) * tau0 * p0)
        C(i) = min(C(i), C_max)
        h(i) = cell_volume(i) ** (1/3)          ! 3-D definition
        Pi_sigma(i) = c_sigma * sigma / h(i) * B(alpha_star(i))
    else:
        C(i) = 0
        Pi_sigma(i) = 0

! p_n is the existing pressure used in the velocity predictor
u_star = velocity_predictor_using(p_n)

! Capillary-biased signed compliance: one linear correction solve
assemble [-div(K grad) + C] phi
        = -div(u_star) - C*(p_n - p_v + Pi_sigma)
solve for phi

! For strict one-sided compliance instead, use an active-set iteration
! with active(i) = trapped(i) and p_n(i) + phi_iter(i) > p_v.

u_new = u_star - K * grad(phi)

```

```

p_new = p_n + phi
update VOF with pressure-corrected face fluxes
verify net liquid inflow = loss of void

```

References

- [1] Computational Physics and Methods Group, *Truchas Reference Manual*, Version 21.01.1, Los Alamos National Laboratory, 2021. <https://www.truchas.org/ReferenceManual.pdf>.
- [2] Truchas developers, *Truchas source repository*, Los Alamos National Laboratory. <https://gitlab.com/truchas/truchas>. Exact commit used for the source-level reconstruction remains to be recorded.
- [3] C. W. Hirt and B. D. Nichols, “Volume of Fluid (VOF) Method for the Dynamics of Free Boundaries,” *Journal of Computational Physics*, 39(1), 201–225, 1981. doi:10.1016/0021-9991(81)90145-5.
- [4] J. B. Bell, P. Colella, and H. M. Glaz, “A Second-Order Projection Method for the Incompressible Navier–Stokes Equations,” *Journal of Computational Physics*, 85(2), 257–283, 1989. doi:10.1016/0021-9991(89)90151-4.
- [5] Lord Rayleigh, “On the Pressure Developed in a Liquid during the Collapse of a Spherical Cavity,” *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science*, 34(200), 94–98, 1917. doi:10.1080/14786440808635681.
- [6] C. E. Brennen, *Cavitation and Bubble Dynamics*, Oxford University Press, 1995.
- [7] J. Li and P. M. Carrica, “An Approach to Couple Velocity/Pressure/Void Fraction in Two-Phase Flows with Incompressible Liquid and Compressible Bubbles,” *International Journal of Multiphase Flow*, 102, 77–94, 2018. doi:10.1016/j.ijmultiphaseflow.2018.01.021.
- [8] S. Shin, “Simulation of Compressibility of Entrapped Air in an Incompressible Free Surface Flow Using a Pressure-Based Method for Unified Equations,” *International Journal for Numerical Methods in Fluids*, 92(10), 1274–1289, 2020. doi:10.1002/fld.4827.
- [9] J. U. Brackbill, D. B. Kothe, and C. Zemach, “A Continuum Method for Modeling Surface Tension,” *Journal of Computational Physics*, 100(2), 335–354, 1992. doi:10.1016/0021-9991(92)90240-Y.